

Tarana Sharmin

Los Angeles, CA | (321)-419-3215 | Taranasharmin07@gmail.com | [linkedin.com/tarana-sharmin](https://www.linkedin.com/tarana-sharmin) | [GitHub: github.com/TaranaSharmIn](https://github.com/TaranaSharmIn)

EDUCATION

University of Southern California

May 2029

B.S. in Computer Science (Games), Viterbi School of Engineering

Minor in Dance, Kaufman School of Dance

- QuestBridge National Match finalist, Deans List
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TECHNICAL SKILLS

Languages: Python, C++, C#, Java, JavaScript

Concepts: Object-oriented programming, data structures, algorithms, data processing, debugging

Libraries & Technologies: Pandas, Three.js, Google Cloud Vision API, Selenium

Developer Tools: Git, GitHub, Perforce, Jira

PROJECTS

UI Engineer | Line by Line, USC Advanced Games Project

July 2026 - Present

- Engineered 4+ gameplay and UI systems in Unity and C#, including the main menu, volume controls, evidence notifications, and centralized UI management, advancing the game toward its production-year milestones.
- Improved clue visibility across 2 player touchpoints by relocating obstructed evidence banners and adding a dynamic journal notification that remains active until players review newly collected evidence.
- Integrated evidence with 3D world-space UI and resolved cross-system interface bugs alongside designers, artists, producers, and engineers using Jira and Perforce..

Gameplay Programmer | Large Language Mimic, Cohort 13 Open Alpha Game Development Club

Jan 2026 - May 2026

- Shipped 2 fully playable minigames in Unity and C#, expanding the game's core narrative satire through interactive gameplay.
- Developed a timed document-redaction system with 3 core components—player input, success conditions, and countdown constraints—and a campaign minigame with dynamic movement and progressive difficulty.
- Integrated and refined features with a 15+ person programming team, using Git and GitHub to debug gameplay, incorporate feedback, and deliver a completed game within a 5-month production cycle.

Software Developer | Select Programming Projects, Remote

June 2023-Present

- **Pinterest Board Analyzer:** Developed a Python and Pandas tool that performs 3 core functions—theme-diversity analysis, day-by-day trend calculation, and automated CSV reporting—converting board data into structured insights.
 - **Tactical RPG Combat System:** Engineered a modular C++ simulation incorporating 5+ interconnected mechanics, including grid collision, Manhattan-distance movement, enemy pathfinding, damage calculations, and persistent status effects.
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EXPERIENCE

Engineering Coach | USC Viterbi First-Year Academy

May 2026 - Present

- Co-instruct weekly ENGR 102 sections and programming workshops for 25–30 first-year engineering students, strengthening technical problem-solving through hands-on activities and progress checks.
- Mentor 25+ students per semester through individualized academic and career guidance, connecting students with 15+ campus resources across advising, tutoring, financial aid, and mental health services.
- Lead 3+ community-building events per semester within a \$250 budget and supported university-wide onboarding for 600+ incoming engineering students, strengthening first-year engagement and belonging.

Laboratory Assistant, NSF CA Barcodes | Natural History Museum of Los Angeles County

August 2026–Present

- Promoted to support a multi-institutional, NSF-funded initiative building a statewide DNA barcode reference library for California's marine invertebrates and algae.
- Manage, preserve, and organize intertidal specimens for integration into the museum's permanent research collections, maintaining accurate taxonomic and physical records.
- Execute quality-control checks across specimen materials and project workflows, identifying organizational errors before records advance into active genetic processing.

Research and Collections Digitizer, NSF DigIn | Natural History Museum of Los Angeles County

August 2026–Present

- Digitized and standardized 300+ historical marine-invertebrate records for an NSF-funded open database, expanding reliable collection access for researchers and the public.
 - Processed 50–200 specimens per shift by validating taxonomic, geographic, and collection metadata against WoRMS and correcting incomplete or inconsistent records.
 - Improved data accuracy and processing efficiency by developing validation scripts, resolving skipped specimen records, and sustaining remote transcription through the grant's final year.
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INVOLVEMENT & LEADERSHIP

Programming Mentor | Open Alpha - Mentor 3–5 new programmers by coordinating tasks, supporting Unity development, and onboarding members to Git workflows.

VR Initiative Member | Shift SC - Explore VR development and its social and ethical implications while collaborating toward a technical VR project.